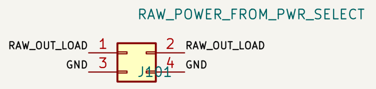
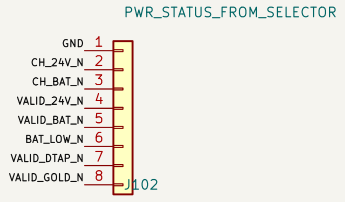
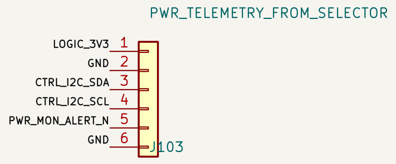
A. PWR–SELECT INTERFACE





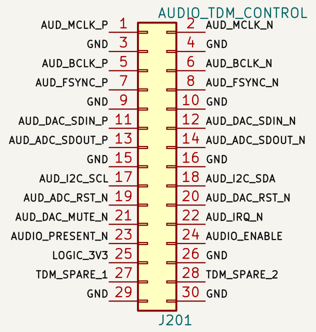


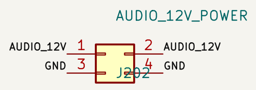
Mates with PWR–SELECT J301; two contacts per polarity; validate 15 A derating.

Mates with PWR–SELECT J401; add 3.3 V pull–ups on the carrier.

Mates with PWR–SELECT J402; carrier supplies 3.3 V and owns I2C pull–ups.

B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL



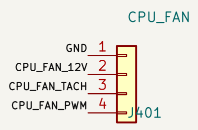


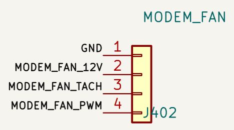
Separate filtered audio–power harness; no fan/radio loads.

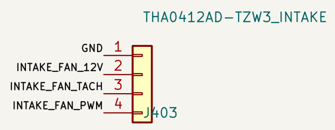
Mates with AUDIO–8X8 J101. Final connector MPN follows 100 ohm pair and vibration review.

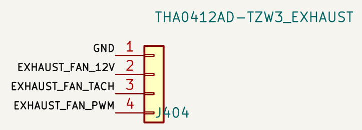
Carrier drives MCLK/BCLK/FSYNC/DAC data; AUDIO–8X8 drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS









CPU_FAN

MODEM_FAN

THA0412AD–TZW3_INTAKE

THA0412AD–TZW3_EXHAUST

J403 is physically keyed/labeled INTAKE; J404 is keyed/labeled EXHAUST.

D. DETAILED CAPTURE SUITE – REV A1

01 CM5–Core–Allocated: exact 300–contact CM5 symbol, 76 locked functions

02 CM5–Core–Allocated: VCC_SYSIN, reset, recovery, power–on and debug UART

03 Network–PCIe: PI7C9X2G608GP, 3x LAN7430 and native WAN1 front end

04 Network–PCIe: 4x Wurth MagJack and Molex 0679101002 Mini PCIe 4T4R Wi–Fi

05 WWAN–SIM: TE 2199230–3 B–key, USB3/USB2, controls and dual–SIM mux

06 Display–Harness: HDMI, USB touch and dedicated 12 V / 2.5 A branch

07 Audio–Control: I2S0 TDM, I2S1 ES8316 headset and CTIA amplifier path

08 Thermal–IO: I2C translation, GPIO expansion, sensors and four fan channels

09 Power–Regulators–A1: protected raw hold–up, all main rails, sequencing and test points

10 Audio–8x8: all XLR shields bonded to chassis with one controlled AGND bond

11 Review: every detailed sheet has zero ERC errors; warnings are classified

DC/DC compensation and magnetics remain a separate power–design calculation milestone.

High–speed routing starts only after stackup and connector Z datums are released.

The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.

NOT A FABRICATION RELEASE
Program audio crosses a buffered differential TDM link
Fan 1 intake, fan 2 exhaust; independent PWM/tach channels
Interface overview; detailed A1 capture is maintained in the named standalone sheets
ProComm

Sheet: /
File: CM5–Carrier.kicad_sch

Title: Radxa CM5 ProComm Carrier – Interface Contract

Size: A2	Date: 2026–08–14	**Rev: A1**
KiCad E.D.A. 10.0.5–rc1		Id: 1/1